

Study Questions: Action

1. Define **Action** in your own words, specifically as it applies to High School.
2. How does the Free Energy Principle constrain our understanding of Action?
3. Contrast the Classical view of Action with the Active Inference view.
4. Explain how gradient descent works as an action policy. If the free energy function is $F(x) = (x - 3)^2$, compute the gradient at $x = 5$ and describe which direction the agent should move.
5. In a PID controller for a thermostat, what do the proportional, integral, and derivative terms each contribute to the action output? Write simplified equations for each term.
6. What is a learning rate in gradient descent, and how does it affect the agent's action? What happens if the learning rate is too large? Too small?
7. Describe the difference between action as "stimulus-response" (classical view) and action as "prediction fulfillment" (Active Inference view) using mathematical notation.
8. A thermostat has a setpoint of 72 degrees F and the current temperature is 65 degrees F. Calculate the proportional control action if the gain constant $K_p = 2.0$.
9. How does the gradient of free energy determine the direction of an agent's action? Explain why actions always follow the steepest descent direction.
10. Compare the mathematical structure of gradient descent in machine learning with the PID controller in control theory. What do they have in common?
11. An agent can take actions that change its environment (active inference) or change its internal model (perceptual inference). Give a mathematical scenario where each strategy is more effective.

12. In calculus, explain why a local minimum of the free energy function might not be the global minimum. How does this relate to an agent getting "stuck" in suboptimal behavior?
13. Write a difference equation that describes how an agent's state changes over time when it takes gradient descent actions: $x(t+1) = x(t) - \alpha * dF/dx$. Simulate three steps starting from $x(0) = 10$ with $F(x) = x^2$ and $\alpha = 0.1$.
14. How does the concept of a control variable in a differential equation relate to the action variable in Active Inference? Provide a concrete example.
15. A robot arm must move to a target position. Formulate this as a free energy minimization problem where the action is the motor command and the prediction error is the distance from the target.
16. Explain why Active Inference predicts that organisms act to confirm their predictions rather than simply respond to stimuli. What mathematical advantage does this provide?
17. What is the relationship between the magnitude of prediction error and the strength of the resulting action? Is this relationship always linear? Explain with a mathematical example.
18. In the PID controller, the integral term accumulates past errors over time. How does this relate to an Active Inference agent that learns from its history of prediction errors?
19. Describe a scenario where taking no action is the optimal action from a free energy perspective. Under what mathematical conditions does this occur?
20. Design a simple control system (not a thermostat) that uses the Active Inference framework. Specify the state variable, the action variable, the prediction error, and the update rule. Calculate the system's behavior over 5 time steps.